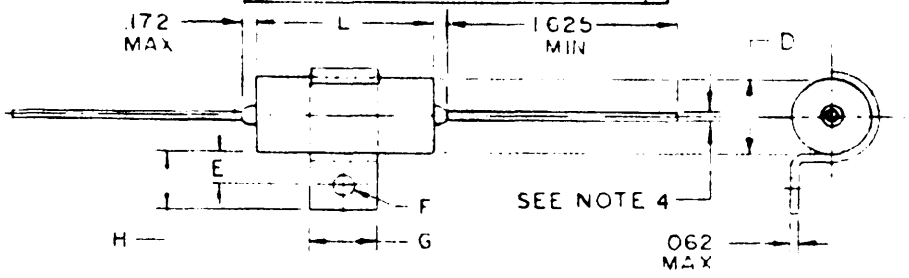
MILITARY SPECIFICATION SHEET

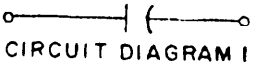
CAPACITORS, FIXED, METALLIZED (PAPER, PAPER PLASTIC, OR PLASTIC FILM) DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES). STYLE CH12

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-18312.

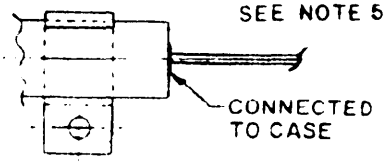
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN AFTER 8 JANUARY 1970. USE MIL-C-39022/8

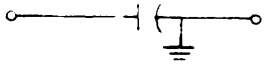




CIRCUIT DIAGRAM 1



INCHES	MM
.062	1.57
.172	4.37
1.625	41.28



CIRCUIT DIAGRAM 3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. See table I for additional dimensions.
4. Number 22 AWG wire for case diameters of .312 (7.92 mm) and under. Number 20 AWG wire for case diameters over .312 (7.92 mm).
5. Bracket wraparound is approximately 135° for case diameter of .400 inch and smaller, and 225° for case diameters greater than .400 inch.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Seal:
Method 112 of MIL-STD-202, condition A.
Liquid-filled capacitors - One hour at applicable high ambient temperature ±3° C with no evidence of leakage.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated dc voltage.
Terminal to case (where case not a terminal) - 200 percent of rated dc voltage.

Capacitance:
Method 305 of MIL-STD-202 (for value and tolerance see table I).

Dissipation factor:
At 25° C - 1.0 percent.
At high test temperature - 2.0 percent.

Insulation resistance:
Terminal to terminal - See table III.
Terminal to case (case not a terminal) - 10,000 megohms.

Barometric pressure:
Method 105 of MIL-STD-202.
Test condition - 0.82 inches of mercury (80,000 ft).

Vibration, high frequency:
Method 204 of MIL-STD-202, condition B.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except that for step 3, temperature shall be 125° C.

Immersion:
Method 104 of MIL-STD-202, condition C.
Insulation resistance:
Terminal to terminal - Not less than 1/3 the value specified in table III.
Terminal to case (case not a terminal) - 5,000 megohms, minimum.
Dielectric withstanding voltage:
Terminal to terminal - 150 percent of rated dc voltage.
Terminal to case (case not a terminal) - 200 percent of rated dc voltage.
Capacitance change - Not to exceed 10 percent of initial value.
Dissipation factor - Not to exceed 110 percent of initial requirement.

Solderability:
Method 208 of MIL-STD-202.

Terminal strength:
Method 211 of MIL-STD-202, conditions A, and D.
Condition A - Applied force 4-1/2 pounds.
Condition D - 3 twists of 360 degrees.

Low temperature and capacitance change with temperature:
Conditions - 48 hours with rated voltage applied.
Capacitance change - See table III.

Life:
Qualification - Test voltage 140 percent of rated voltage, 1,000 hours at high ambient temperature.
Quality conformance - 250 hours at conditions specified for 1,000-hour test.

Dissipation factor:
At high ambient test temperature (any time 24 to 48 hours from start of test) - Not greater than 2.0 percent.
At high ambient test temperature (anytime during last 48 hours of test) - Not greater than 2.5 percent.
At 25° C (after life test) - Not greater than 1.25 percent.

Capacitance:
At 25° C (after life test) - Within 10 percent of initial value.

Insulation resistance:
At 25° C (after life test) - Terminal to terminal - Not less than 1/3 value specified in table III.
Terminal to case (case not a terminal) - Not less than 5,000 megohms.

Supersession information: See table I.

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TABLE I. Dimensions, and dash numbers.

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches)						Superseded MIL-C-18312 3A dash numbers	
				L ±.062	D ±.031	H ±.062	G ±.062	E ±.031	F ±.005	Capacitance tolerance ±10%	Capacitance tolerance -20%
	Volts	µf									
CH12A1NC104-	200	0.10	1	0.844	0.312	0.312	0.250	0.188	0.144	0201	0202
CH12A3NC104-	200	0.10	3	0.781	0.312	0.312	0.250	0.188	0.144	0203	0204
CH12A1NC224-	200	0.22	1	1.125	0.312	0.312	0.250	0.188	0.144	0205	0206
CH12A3NC224-	200	0.22	3	1.062	0.312	0.312	0.250	0.188	0.144	0207	0208
CH12A1NC474-	200	0.47	1	1.125	0.400	0.312	0.250	0.188	0.144	0209	0210
CH12A3NC474-	200	0.47	3	1.062	0.400	0.312	0.250	0.188	0.144	0211	0212
CH12A1NC105-	200	1.0	1	1.125	0.562	0.438	0.500	0.250	0.156	0213	0214
CH12A3NC105-	200	1.0	3	1.062	0.562	0.438	0.500	0.250	0.156	0215	0216
CH12A1NC155-	200	1.5	1	1.844	0.562	0.438	0.500	0.250	0.156	0217	0218
CH12A3NC155-	200	1.5	3	1.781	0.562	0.438	0.500	0.250	0.156	0219	0220
CH12A1NC225-	200	2.2	1	1.844	0.562	0.438	0.500	0.250	0.156	0221	0222
CH12A3NC225-	200	2.2	3	1.781	0.562	0.438	0.500	0.250	0.156	0223	0224
CH12A1NC335-	200	3.3	1	1.875	0.670	0.438	0.500	0.250	0.156	0225	0226
CH12A3NC335-	200	3.3	3	1.812	0.670	0.438	0.500	0.250	0.156	0227	0228
CH12A1NC475-	200	4.7	1	1.875	1.000	0.438	0.500	0.250	0.156	0229	0230
CH12A3NC475-	200	4.7	3	1.812	1.000	0.438	0.500	0.250	0.156	0231	0232
CH12A1NC685-	200	6.8	1	1.875	1.000	0.438	0.500	0.250	0.156	0233	0234

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches)						Superseded MIL-C-18312/3A dash numbers	
				L ±.062	D ±.031	H ±.062	G ±.062	E ±.031	F ±.005	Capacitance tolerance ±10%	±20%
	Volts	µf									
CH12A3NC685-	200	6.8	3	1.812	1.000	0.438	0.500	0.250	0.156	0235	0236
CH12A1NC106-	200	10.0	1	2.375	1.000	0.438	0.500	0.250	0.156	0237	0238
CH12A3NC106-	200	10.0	3	2.312	1.000	0.438	0.500	0.250	0.156	0239	0240
CH12A1NC126-	200	12.0	1	2.625	1.000	0.438	0.500	0.250	0.156	0241	0242
CH12A3NC126-	200	12.0	3	2.562	1.000	0.438	0.500	0.250	0.156	0243	0244
CH12A1NE473-	400	0.047	1	1.125	0.312	0.312	0.250	0.188	0.144	0245	0246
CH12A3NE473-	400	0.047	3	1.062	0.312	0.312	0.250	0.188	0.144	0247	0248
CH12A1NE104-	400	0.10	1	1.125	0.400	0.312	0.250	0.188	0.144	0249	0250
CH12A3NE104-	400	0.10	3	1.062	0.400	0.312	0.250	0.188	0.144	0251	0252
CH12A1NE224-	400	0.22	1	1.125	0.562	0.438	0.500	0.250	0.156	0253	0254
CH12A3NE224-	400	0.22	3	1.062	0.562	0.438	0.500	0.250	0.156	0255	0256
CH12A1NE474-	400	0.47	1	1.625	0.562	0.438	0.500	0.250	0.156	0257	0258
CH12A3NE474-	400	0.47	3	1.562	0.562	0.438	0.500	0.250	0.156	0259	0260
CH12A1NE105-	400	1.0	1	1.875	0.750	0.438	0.500	0.250	0.156	0261	0262
CH12A3NE105-	400	1.0	3	1.812	0.750	0.438	0.500	0.250	0.156	0263	0264
CH12A1NE225-	400	2.2	1	2.125	1.000	0.438	0.500	0.250	0.156	0265	0266
CH12A3NE225-	400	2.2	3	2.062	1.000	0.438	0.500	0.250	0.156	0267	0268

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches)						Superseded MIL-C-18312/3A dash numbers	
				L ±.062	D ±.031	H ±.062	G ±.062	E ±.031	F ±.005	Capacitance tolerance ±10%	±20%
	Volts	µf									
CH12A1NF335-	400	3.3	1	2.625	1.000	0.438	0.500	0.250	0.156	0269	0270
CH12A3NF335-	400	3.3	3	2.562	1.000	0.438	0.500	0.250	0.156	0271	0272
CH12A1NF103-	600	0.01	1	0.812	0.312	0.312	0.250	0.188	0.144	0273	0274
CH12A3NF103-	600	0.01	3	0.750	0.312	0.312	0.250	0.188	0.144	0275	0276
CH12A1NF223-	600	0.022	1	0.812	0.312	0.312	0.250	0.188	0.144	0277	0278
CH12A3NF223-	600	0.022	3	0.750	0.312	0.312	0.250	0.188	0.144	0279	0280
CH12A1NF473-	600	0.047	1	1.125	0.400	0.312	0.250	0.188	0.144	0281	0282
CH12A3NF473-	600	0.047	3	1.062	0.400	0.312	0.250	0.188	0.144	0283	0284
CH12A1NF104-	600	0.10	1	1.125	0.500	0.312	0.250	0.188	0.144	0285	0286
CH12A3NF104-	600	0.10	3	1.062	0.500	0.312	0.250	0.188	0.144	0287	0288
CH12A1NF224-	600	0.22	1	1.375	0.562	0.438	0.500	0.250	0.156	0289	0290
CH12A3NF224-	600	0.22	3	1.312	0.562	0.438	0.500	0.250	0.156	0291	0292
CH12A1NF474-	600	0.47	1	1.625	0.670	0.438	0.500	0.250	0.156	0293	0294
CH12A3NF474-	600	0.47	3	1.562	0.670	0.438	0.500	0.250	0.156	0295	0296
CH12A1NF105-	600	1.0	1	1.844	1.000	0.438	0.500	0.250	0.156	0297	0298
CH12A3NF105-	600	1.0	3	1.781	1.000	0.438	0.500	0.250	0.156	0299	0300
CH12A1NF155-	600	1.5	1	1.875	1.000	0.438	0.500	0.250	0.156	0301	0302

See footnote at end of table I.

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TABLE I. Dimensions, and dash numbers. - Continued

Type designation 1/	DC rated voltage	Capacitance value	Circuit diagram	Dimensions, nominal (inches)						Superseded MIL-C-18312/3A dash numbers	
				L ±.062	D ±.031	H ±.062	G ±.062	E ±.031	F ±.005	Capacitance tolerance ±10%	±20%
	Volts	µf									
CH12A3NF155-	600	1.5	3	1.812	1.000	0.438	0.500	0.250	0.156	0303	0304
CH12A1NF225-	600	2.2	1	2.625	1.000	0.438	0.500	0.250	0.156	0305	0306
CH12A3NF225-	600	2.2	3	2.562	1.000	0.438	0.500	0.250	0.156	0307	0308

1/ Complete type designation will include an additional symbol for capacitance tolerance.

TABLE II. Millimeter equivalents of decimal inches.

Inches	MM	Inches	MM	Inches	MM	Inches	MM
.005	0.13	.400	10.16	.844	21.44	1.812	46.02
.031	0.79	.438	11.13	1.000	25.40	1.844	46.84
.062	1.57	.500	12.70	1.062	26.97	1.875	47.63
.144	3.66	.562	14.27	1.125	28.58	2.062	52.37
.156	3.96	.670	17.02	1.312	33.32	2.125	53.98
.188	4.78	.750	19.05	1.375	34.93	2.312	58.72
.250	6.38	.781	19.84	1.562	39.67	2.375	60.33
.312	7.92	.812	20.62	1.625	41.28	2.562	65.07
				1.781	45.24	2.625	66.68

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TABLE III. Characteristic.

Operating temperature range	-55 to 125 ±3° C	
DC voltage rating (volts)	200	400 600
High ambient test temperature	125° ±3° C	125° ±3° C
Low ambient test temperature	-55° ±3° C	-55° ±3°C
Megohms x microfarads:		
At 25° C	2,000 min	2,000 min
At high ambient test temperature	10 min	40 min
Insulation resistance (megohms):		
At 25° C 1/	12,000	12,000
At high ambient test temperature 1/	150	600
Capacitance change at low temperature	-15% max	-15% max
Life test voltage (% rated voltage)	140%	140%

1/ Need not exceed.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85

User activities:
Army - None
Navy - AS, OS, MC, CG
Air Force - 14, 19

Preparing activity:
Navy - EC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

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NAVAL ELECTRONIC SYSTEMS COMMAND
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